



Research Paper

Optimal techno-economic potential and site evaluation for solar PV and CSP systems in Ghana. A geospatial AHP multi-criteria approach

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ABSTRACT

In this study, the techno-economic potential and site suitability for utility-scale solar PV and CSP technologies have been estimated for Ghana. The overarching goal is to increase visibility of solar energy potentials to encourage and facilitate the uptake of renewable energy towards the global decarbonisation agenda as well as to expand energy access in the country. To assist policymakers in Ghana towards this agenda including potential areas for investment, the geographical, technical, and economic potentials for the two solar energy technology solutions have been assessed. In this study unlike many other studies, the AHP and Weighted Sum Average approaches have been applied in the ArcGIS Pro environment to estimate the two technology potentials at the national and regional levels. This combined approach is still in its infancy stages and thus contributes to a new body of knowledge from the international perspective in addition to its applicability at specific country level like Ghana. Results from the analysis show varying generation capacities for the selected solar energy technologies. Geographically, about 85% of total land area in Ghana is suitable for solar energy deployment. At the national level, a generation capacity of ~68,622 TWh/year can be harnessed from utility-scale PV and ~23,452 TWh/year from CSP systems. The LCOE for the utility-scale solar PV technology ranges from a minimum of about \$0.04/kWh to a maximum of \$0.15/kWh, and from ~\$0.73/kWh to ~\$2.89/kWh for the CSP technology in Ghana.

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Introduction

Access to energy or electricity is a basic need of every human being as stated in UN Agenda 2030 and central to achieving the Sustainable Energy for All (SE4ALL) initiative with focus on SDG 7 [1–4]. Access to energy is an enabler for sustainable development and thus, plays an integral role in the development of a country. It is one of the major economic boosters and development indicators of a country. According to the World Bank and Sustainable Energy for All organizations, about 17% of the Ghanaian population lived without electricity access in 2019 [5]. Energy consumption in Ghana on the average is 50–100 times lower than in developed countries [6,7]. For instance, in 2018, annual electricity consumption per capita in Ghana was only 0.3 MWh/capita compared to in developed countries like Germany with 6.9 MWh/capita, and 13.1 MWh/capita in the USA [8]. Even though there is no universally accepted threshold of the minimum accepted consumption per capita value, experts like Javadi, et al. [9] posit that, an assessment between basic quality of life, basic survival, internal collaboration and social graces, “annual electricity consumption of about

1000 kWh per year per capita is known as the boundary between basic life condition and guaranteed survival” [9,10].

Efforts by the government of Ghana towards increasing electricity access dates to the 1960s, when the then government developed one of the country's ambitious electrification plans through the development of hydroelectric power projects. This led to the development of the Akosombo Dam across the Volta River [11,12] the major hydro-electric power plant in the country. Following, the country saw an increase in the development of two additional mini-hydro power plants, the Kpong and Bui hydro power stations, making hydropower the second major share and source of clean energy in the country's energy generation mix. This accounted for about 40% of total installed generation capacity of which Akosombo supplied about 1020 MW; Kpong 160 MW, and Bui 40 MW as at 2018 [12]. Also, as of 2018, grid capacity was about 14,067 GWh of which 39.2% (i.e., 4991 GWh) came from hydropower.

However, currently with the impact of climate change, the capacity of hydro has significantly reduced because of shortage of water inflow into the dams. Consequently, with an existing and increasing demand, the country was compelled to find additional power sources to bridge the huge demand and supply gap. The government then resorted to fossil fuels in its quest to service the unmet energy demands in the country thereby increasing the share and making thermal power the major source of electricity

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Nomenclature

AHP	Analytic Hierarchy Process	MCDM	Multi-criteria decision-making
CI	Consistency Index	MW	Megawatt
CR	Consistency Ratio	MWh	Megawatt hour
CSP	Concentrated Solar Power	NDC	Nationally Determined Contribution
DNI	Direct Normal Irradiance	O&M costs	Operations and Maintenance costs
ELECTRE	ELimination Et Choix Traduisant la REalité	PROMETHEE	Preference Ranking Organization METHod for Enrichment of Evaluations
EU	European Union	PR _{pv}	Performance Ratio of photovoltaic module
FAHP	Fuzzy Analytic Hierarchy Process	RE	Renewable energy
GHI	Global Horizontal Irradiance	RI	Random Index
GIS	Geographic Information Systems	SAM	System Advisor Model
GSA	Global Solar Atlas	SDG 7	Sustainable Development Goal 7
GWh	Gigawatt hour	SE4ALL	Sustainable Energy for All
GWh/y	Gigawatt hour per year	Solar PV	Solar Photovoltaics
IRENA	International Renewable Energy Agency	TOPSIS	Technique for Order of Preference by Similarity to Ideal Solution
IUCN	International Union for Conservation of Nature	TWh/year	Terawatt hour per year
Km	Kilometer	UN	United Nations
km ²	Square kilometer	UNEP	United Nations Environment Programme
kW	Kilowatt	WCMC	World Conservation Monitoring Centre
kWh	Kilowatt hour	\$	United States Dollar
kWh/m ² /year	Kilowatt hour per square meter per year		
LCOE	Levelized cost of electricity		

supply in the country [1,13–15]. As at 2020, thermal power accounted for about 67.91% of grid electricity supply in the country [13], hydro generation at 31.79% of hydro; with still insignificant share of 0.30% from solar photovoltaics (PV) [13]. More so, even with the additional supply, the country is still unable to meet the basic electricity needs of the people.

Further, with increasing concern about the use of conventional energy sources contributing to greenhouse gas emissions and an environmental pollutant, the use of renewable energy (RE) technologies like solar, wind, hydro, etc., have gained prominence and political attention. Also, without global transitioning of the energy sector, it will be impossible to achieve the 1.5 degree target of the Paris Agreement [16]. More significantly, also because renewable energy generation costs have experienced a sharp fall over the past decade due to economies of scale, technology improvement, growing developer experiences as well as a growing competitive supply chain in the RE industry globally [17–18] making RE a more competitive alternative energy source. Again, according to IRENA [17], renewable energy technologies have become the least-cost option for alternative supply in almost all parts of the world with about 72% increasing deployment in 2019. Levelized cost of electricity (LCOE) for utility-scale solar PV fell 82% between 2010 and 2019, whereas that of concentrated solar power (CSP) fell 47% as reported by IRENA [17]. The call to diversify the country's energy system using RE resources has even become more crucial for three main reasons; i) to assist the government of Ghana meet its sustainable energy for all Nationally Determined Contribution (NDC) targets by 2030 [1], ii) in fulfilment of its Paris Agreement commitments to contribute and reduce its greenhouse emission, iii) and help cut-down investment costs accrued in importing crude oil to fuel existing thermal plants which will eventually assist government achieve its debt recovery in the energy sector [19,20].

Just like many RE resources, solar energy development involves strong geospatial implication [21]. Solar radiation intensities vary considerably from one location to another. For instance, according to IRENA's CSP technology brief, unlike solar PV systems, CSP plants require high direct solar irradiance for the plants to work and as such one of the major reasons places in the Sun Belt region (i.e. between 40 degrees north and south of the equator) show good potential for CSP deployment [22]. This implies that the selec-

tion of a suitable site for solar energy planning requires consideration for certain fundamental conditions that the chosen site must fulfil to ensure optimal solar energy generation. These criteria may include but not limited to, proximity to the grid infrastructure for large scale or utility grid-connected systems as well as considerations to how to provide cost-effective options for the end-user while minimizing the overall production costs of the project and maximizing energy output from the solar technology used [23].

The geographical location of Ghana provides the country with an excellent potential for solar energy development [24]. And thus, performing a thorough solar energy site assessment is a strategic step towards ensuring a well-performing and cost-effective solar energy project [23]. Furthermore, as mentioned earlier, the geographical characteristics of solar irradiance has led to a growing importance in geospatial-based studies in estimating the solar energy potential of a location and help determine a more viable solution to service both urban and off-grid or rural households [25]. Moreover, the development of many solar technologies for a site is mostly backed by policy programs and decisions to assist policymakers make a well-informed decision towards their sustainable energy access interventions and programmes. Multi-criteria decision-making (MCDM) methods have helped to make decisions when confronted with multiple and conflicting objectives [26]. More so, to assist policymakers achieve this goal, several geographic information systems (GIS) based multi-criteria decision-making approaches have been applied to help determine the best site for the development of a specific type of solar technology; in this case for utility-scale PV and CSP technologies.

The application of GIS-based MCDM method has been used in several studies to assess potential sites for solar energy projects. For instance, Wang, Jing et al. [27]; Kumar, Sah et al. [28]; and Choi, Suh et al. [29] have provided an extensive review of different MCDM methods for RE resource mapping and planning across different geographies. Some methods highlighted included the Analytic Hierarchy Process (AHP) by Saaty [30], FAHP by Zadeh [31], Weighted Sum Model by Fishburn [32], Weighted Average by Anderson [33], Boolean Overlay [34], Weighted Product Model by Bridgman [35]; ELECTRE by Benayoun et al. [36], TOPSIS by Hwang and Yoon [37]; Multi-Attribute Utility Technique by Edwards and Newman [38], PROMETHEE by Brans and Vincke [39], and Hierarchical Cluster Analysis [40], etc. These methods have been applied

in various studies in many countries such as South Africa [41,42]; in Turkey [43], Croatia [44], West Africa [7], Canada [45], USA [46], just to mention a few.

Despite all the MCDM techniques, the most widely used is the AHP technique developed by Thomas, L. Saaty [30]. However, despite the usefulness of the AHP method, it is not comprehensive on its own without the application of additional decision-making methods [28]. For instance, the AHP approach alone does not address the geographical conditions of the location. Many studies have combined AHP with other methods like the fuzzy logic, Goal Programming etc., to reduce the complexity when applied to a number of criteria [47]. The application of AHP method combined with GIS techniques have been applied in many solar energy evaluation studies, for example, in Saudi Arabia [23,48], in Indonesia [49,50], in China [51], Morocco [52], Brazil [53], Japan and Malaysia [54] etc. Colak, Memisoglu et al. [43] applied GIS and the AHP methods to select optimal site for solar PV power plants in the Malatya Province in Turkey. The GIS-AHP approach has also been applied to explore the potential of CSP in Zimbabwe [55], and solar energy potential in the Bumthang Valley in Bhutan [56]. Yushchenko et al. [7] also carried out a GIS-based evaluation of solar PV and CSP potential in the West Africa region which included Ghana. In the present study unlike many other studies, the AHP and Weighted Sum Average approaches have been applied; this combined approach is still in its infancy stages and thus contributes to a new body of knowledge from the international perspective in addition to its applicability at specific country levels. Further, a review of the literature shows that, this study and approach is the first of its kind in Ghana providing and considering a multi-criteria and multi-dimensional approaches by taking into account the national electrification policy which is to extend grid connection to areas within a 20km distance with a population density of 500persons/ km² and above who live within the 20km existing grid buffer distance [57,58].

In addition, one key aspect missing in most of the studies mentioned above is an assessment of the financial viability of the solar project at specific locations. To address this, the LCOE method which is a measure of the lifetime costs of generating electricity from energy technologies has been used by many experts [59–61] to model the potential costs involved in embarking on solar projects. Researches done by the EU solar bankability project [60] and, IRENA [61] have used and established a common practice using the LCOE method to minimize the risks associated with investments in solar PV and CSP projects. In a techno-economic feasibility study on CSP by Aly, Bernardos et al., [69] in Tanzania, they reiterated that the penetration of CSP projects in Tanzania is however dependent on the associated financial implications. Furthermore, even in studies where the LCOE method has been used to compute the life-cycle costs of solar project, an assessment of the geographical viability is usually not included or evaluated. Many of the solar energy assessments are based on an either-or approach, i.e., it is either purely a techno-economic assessment or just an evaluation of the geographical or technical potential like as seen in the studies mentioned above. In addition, there has also been recent and increasing interest by policy makers and investors to have a visualization and knowledge of specific areas that would yield the optimum returns and consumer satisfaction. In this present study, both the techno-economic and evaluation of the sites have been analyzed to provide a more comprehensive insight into potential areas of interest for investment.

In Ghana, geospatial techno-economic and site assessments of the two selected solar energy technologies are still in their infancy and limited, not to mention an assessment for all regions. For instance, in a study by Agyekum and Velkin [62], the System Advisor Model (SAM) software was used to assess the techno-economic viability of CSP for only two selected towns in the northern part of Ghana. Their assessment was however only limited to CSP, two

towns as well as the study did not consider an evaluation of the site to assess its viability for the CSP project. A review of the literature also show studies carried out by IRENA to estimate the potential of renewable energy in the entire Africa region in 2014 [63]. Results from IRENA' analysis show that, about 229 TWh/year of CSP and 7644 TWh/year of large-scale PV can be generated in Ghana based on their selected suitability criteria [63,64]. IRENA's assessment did not also factor in the economic viability of solar project. The concept of evaluating the techno-economic and site feasibility of solar energy potential using geospatial tools and software is still in its early stages. This makes the current study a maiden effort towards opening-up further discussions and studies on the development of solar energy markets and investment risks, even at more disaggregate levels. An assessment at disaggregate levels would also promote and encourage decentralization of electrification planning at the local levels.

Research objective

The main objective of this study is to ascertain the feasibility of utility-scale solar PV, and parabolic trough CSP systems penetration in Ghana by assessing the techno-economic and conduciveness of potential locations for the technology deployment. For this purpose, the geographical, the technical and the economic viability of utility-scale solar PV, and parabolic trough CSP systems have been evaluated using GIS and AHP multi-criteria methods.

Methodology

This chapter of the study provides detail description of the methods used and applicability using Ghana as a case study. In this study, the techno-economic potentials and site suitability for utility-scale solar PV and parabolic trough CSP technologies have been estimated. Despite the differences among the CSP technologies, the parabolic trough dominates the CSP markets and is the most commercially matured CSP technology [22]. The analysis applied a combined geospatial multi-criteria method, i.e., applied GIS and the AHP multi-criteria analysis methods. The input datasets that were used for this study were all open-sourced and resampled to the same resolution to achieve uniformity in level of detail of study results. All datasets were interpolated to a 1km² grid cell size and the two solar energy potentials were estimated and visualized for all 10 regions in Ghana. It must be mentioned that the site evaluation parameters and values used in this study were not only based on evidences from empirical studies in the Africa region [7,63] but also consideration was given to the national electrification policy of Ghana [58]. Hence, in the present paper, consideration for people living in close proximity to the grid network and with higher population densities were more suitable and assigned higher weights in constructing the pairwise comparison matrices.

Study area

Ghana is located in the western part of Africa at latitude 7.9465° N, and longitude of 1.0232° W. The country shares its borders with Togo to the east, Côte d'Ivoire to the west, Burkina Faso to the north, and to the south, lies the Atlantic Ocean and the Gulf of Guinea. Ghana covers an area of about 238,723 km². The terrain consists mostly of low plains with scattered hills and highlands. The highest point in Ghana, Mt. Afadjato which forms part of the Akwapim-Togo Ranges is ~885m above sea level [65]. The country is also home to the world's largest artificial lake by surface area (~8482 km²): the Lake Volta [65]. The climate in Ghana is tropical: warm and humid with annual mean temperature ranging from about 26–29 °C [66]. Total population as at 2021 is about 31 million with average household size of 3.6 [67].

It is however important to mention that the regional boundaries used for this study were the former 10 regions of Ghana. This is due to the challenges in acquiring GIS administrative boundaries data for the recent 2018 referendum that led to the creation of 6 new regions from the former 10 regions [67,68] as shown in Figure 1.

Concept of the Analytic Hierarchy Process (AHP)

The AHP approach is a multi-criteria decision-making approach that uses hierarchical structures to illustrate a problem and help make judgement based on expert advice to deduce priority scale [30,55]. With AHP, combination of qualitative and quantitative inputs are allowed to help deal with complex problems in energy planning [23]. AHP is used to mitigate the uncertainty and inconsistencies that may arise in the evaluation process [23,29]. A consistency ratio (CR) must also be calculated to determine the degree of inconsistency in the judgment of decisions through the pairwise comparison process. Here all identified criteria are compared against each other in the pairwise comparison matrix and then numerical values are assigned to a criterion expressing judgement of relative importance of a criterion against another [30,69]. With the AHP method, a scale of comparison ranging from values of 1 to 9 is used to describe the intensity/ level of importance among the compared criteria, where the value of 1 is used to represent “equal importance” and 9 for criteria having “extreme

importance” over the other criteria. Table 1 and Table 2 show the pairwise comparison matrix of the five major criteria used for this study which were derived from extensive review of literature to identify suitable areas for utility-scale solar PV and CSP systems in Ghana. To apply this method, first an unstructured problem must be disintegrated in hierarchical order, define the goal, and then identify the alternatives and criteria. Next, a pairwise comparison matrix (**A**) is constructed and **n** as the number of criteria. The matrix depicts the expert judgment of the pair-wise comparisons. Matrix (**A**) will then represent a matrix where each entry a_{ij} of the matrix defines the significance of the i_{th} criterion to the j_{th} criterion [23,30,52]. The below steps provide further insight into carrying out AHP for this study:

$$A = \begin{bmatrix} 1 & a & b \\ 1/a & 1 & c \\ 1/b & 1/c & 1 \end{bmatrix} \quad (1)$$

After computing the pairwise comparison matrix, the next step is to check for the consistency of the matrix. As mentioned earlier, even though the values assigned to criteria are based on expert opinions and experiences and not set arbitrarily, it is still important to determine the level of inconsistencies in the matrix as people's decisions and preferences are sometimes intransitive and inconsistent, and this may cause perturbations in computing the eigenvector of a criteria [69]. Hence, the next step of the AHP

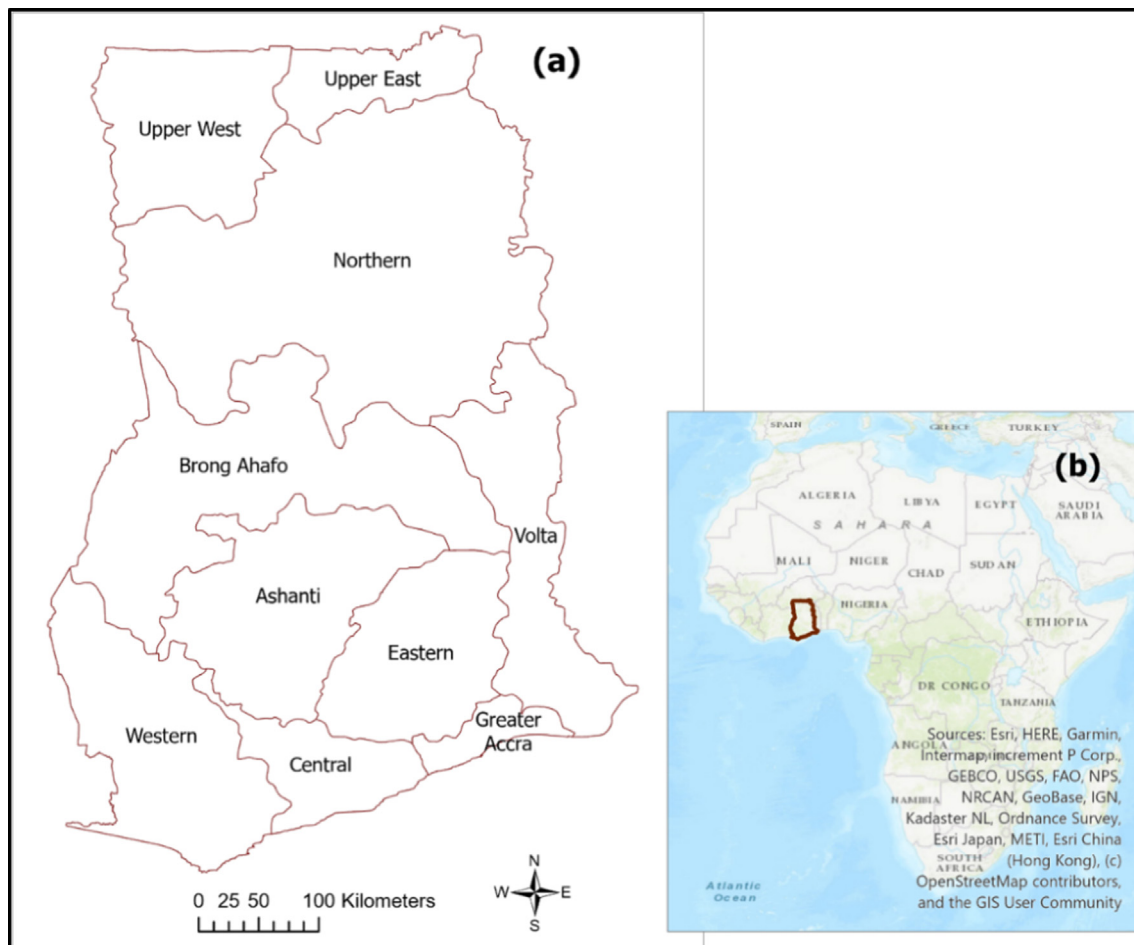


Figure 1. Map of Ghana with the 10 regions. (a) Map of Ghana, (b) Geographical location of Ghana.

Table 1

Pairwise comparison matrix for the utility-scale PV system.

Criteria	Slope	Population Density	Distance to major roads	Solar Irradiance (GHI)	Distance to Electricity Grid Network
Slope	1.00	0.33	0.20	0.13	0.11
Population Density	3.00	1.00	0.33	0.20	0.11
Distance to major roads	5.00	3.00	1.00	0.25	0.13
Solar Irradiance (GHI)	8.00	5.00	4.00	1.00	0.50
Distance to Electricity Grid Network	9.00	9.00	8.00	2.00	1.00

Table 2

Pairwise comparison matrix for the CSP system.

Criteria	Slope	Distance to Rivers	Distance to Electricity Grid Network	Population Density	Solar Irradiance (DNI)
Slope	1.00	0.50	0.50	0.33	0.20
Distance to Rivers	2.00	1.00	0.33	0.50	0.33
Distance to Electricity Grid Network	2.00	3.00	1.00	0.50	0.50
Population Density	3.00	2.00	2.00	1.00	0.50
Solar Irradiance (DNI)	5.00	3.00	2.00	2.00	1.00

Table 3Values for **RI**.

<i>n</i>	2	3	4	5	6	7	8	9
RI	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45

method is to calculate the consistency ratio **CR** which is defined as the ratio of the consistency index **CI** to the average consistency index **RI**¹. **CR** is thus given as [30]:

$$CR = \frac{CI}{RI} \quad (2)$$

The value of **RI** is based on the average of consistency of square matrices of number of observed criteria **n** with its corresponding assigned **RI** value. Table 3 shows the **RI** value of the given number of observations **n** as proposed by Thomas L. Saaty [30].

Next is to compute the consistency index **CI** which is given as [30]:

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (3)$$

If the value for **CR** is < 0.1, then the degree of consistency is acceptable otherwise, decision judgment is inconsistent and based on experiences and expert opinions Saaty [30,70] recommended a revision of the preference pairwise matrix if **CR** > 0.1. Computation of criteria weights resulting from the pairwise matrices for the 2 selected solar energy technologies are shown in Table 4 and Table 5.

Area suitability criteria: Input raster layers

This section highlights all the input variables or decision criteria that were considered in this study for selecting optimal sites for solar energy projects in Ghana (see Figure 2). All input layers were rasterized and reclassified into 5 classes and level of importance was assigned to areas from a scale of 1 to 5, with 1 having the highest prominence.

(a) Solar Irradiance

The first question that generally comes to mind when considering solar energy project is if the location receives enough solar irradiance worth harnessing for electricity. It is one of the main necessary criteria that should be available to determine the suitability of a site for the installation of a solar technology [7,23,29,43,44,52,54–56,63,64,69,71–74]. Solar radiation datasets

for both the Global Horizontal Irradiance (GHI) and the Direct Normal Irradiance (DNI) were openly-sourced from the Global Solar Atlas (GSA) portal [75]. From the nearest resample technique used, GHI values in Ghana ranges from 1 500 kWh/m²/year to 2 080 kWh/m²/year which makes the solar irradiance intensity in Ghana very viable for utility-scale PV system, as well as for residential or a distributed electrification system. The DNI intensity index in Ghana on the other hand is rather nominal; ranging from 700 kWh/m²/year to 1 450 kWh/m²/year which makes CSP-based electrification in Ghana more conducive for micro-grids or distributed systems rather than for large-scale CSP projects.

(b) Slope

The gradient of the topography of the selected site can greatly affect the amount of solar radiation received by the site. The landscape must usually not tilt more than 100% (45°) for PV systems [63,64] and less the 3.7% (2.1°) [63,64] for CSP systems to ensure optimal reception of the solar radiation. Hence prominence was given to areas with lower slope. The slope dataset was accessed from the Land Use Change and Agriculture Program, International Institute for Applied Systems Analysis portal [76].

(c) Population Density

The number of persons per km² of the site is an important factor to consider in determining the optimal technology solution to assign. This particularly helps to identify where demand is to offer best-least cost solutions. The population density dataset used was extracted from the 2020 Africa population by WorldPop [77] in a tiff format. The downloaded data was for the entire Africa continent and the *extract by mask* tool in ArcGIS was used to extract that of Ghana. Areas with higher population densities with given higher prominence.

(d) Grid Infrastructure

As the name suggests, proximity to the power grid plays an integral role in especially assigning an electrification solution to the end-user. It therefore importance to site the solar plant in areas with existing or proposed transmission and distribution networks. This is also to ensure that there is enough demand to meet the energy generated and to offer optimal cost-effective solutions.

¹ The average consistency index **RI** is also referred to as the random index [70].

Table 4

Criteria weights for the utility-scale solar PV system.

Criteria	Slope	Population Density	Distance to major roads	Solar Irradiance (GHI)	Distance to Electricity Grid Network	Weights (%)
Slope	0.04	0.02	0.01	0.03	0.06	3.33
Population Density	0.12	0.05	0.02	0.06	0.06	6.21
Distance to major roads	0.19	0.16	0.07	0.07	0.07	11.35
Solar Irradiance (GHI)	0.31	0.27	0.30	0.28	0.27	28.53
Distance to Electricity Grid Network	0.35	0.49	0.59	0.56	0.54	50.58

Here, **CI** = 0.076 **CR** = 0.068**Table 5**

Criteria weights for the CSP system.

Criteria	Slope	Distance to Rivers	Distance to Electricity Grid Network	Population Density	Solar Irradiance (DNI)	Weights (%)
Slope	0.08	0.05	0.09	0.08	0.08	7.42
Distance to Rivers	0.15	0.11	0.06	0.12	0.13	11.26
Distance to Electricity Grid Network	0.15	0.32	0.17	0.12	0.20	19.08
Population Density	0.23	0.21	0.34	0.23	0.20	24.25
Solar Irradiance (DNI)	0.38	0.32	0.34	0.46	0.39	37.99

Here, **CI** = 0.039 **CR** = 0.034

Also, the further the grid network, the more expensive the LCOE, since the transmission network will have to be extended to the generation point and to where demand is located, thereby increasing overall costs of production. The grid infrastructure dataset was accessed from the World Bank data catalogue portal [78].

(e) Road Network

Conventionally, the location of a solar plant must be easily accessible [52] to allow for easy access during all the development phases; from preliminary site evaluation, plant installation, transmission and for maintenance purposes. Access to road infrastructure goes a long way to help minimize investment costs [7,49,50,79], especially cost associated with transportation. It is therefore of an economic and technical relevance to consider proximity to road infrastructure in evaluating suitable sites for solar electrification projects. In addition, it is expected that, people are more likely to live in areas where they could have access to transport and therefore important to include proximity to road infrastructure in identifying suitable locations for siting a solar plant. Road proximity evaluation also helps to avoid the oversight of installing the power plant in areas designated for road construction. Areas closer to roads were more preferred and assigned higher score of importance.

(f) Rivers

Availability of water is an important factor to consider especially for CSP systems. Therefore, the closer the power plant is to a water body, the more convenient. Just like many conventional thermal power plants, most CSP installations need water to cool and condense the steam cycle [22]. The chosen CSP technology selected must be regularly cleaned off dust that settles on the mirrors to ensure sufficient reception of the solar insolation [80–82]. Hence, areas closer to water bodies were considered more suitable for siting a CSP plant. It must be mentioned that availability of this criterion was only considered for CSP systems. The river dataset was accessed from the World Bank data catalogue portal [83].

(g) Area Restriction Mask

In this study, for ecological and environmental protection reasons, delimited areas like international protected areas were com-

pletely excluded from all the solar project site selection as it is not advisable to install power plants in protected areas [53,84]. Protected areas are one of the major restricted areas considered in this study and they include areas like wetlands, Ramsar sites, forest reserves, parks, etc. The protected areas dataset was openly accessed at UNEP-WCMC and IUCN [85].

Estimation of the geographical potential

The geographical potential refers evaluation of sites for development of the 2 solar technologies. The topography of Ghana's terrain must fulfil certain necessary and minimum environmental conditions to ascertain the viability of installing a solar PV or CSP plant on a site to ensure optimal energy generation and deployment. The evaluation of the geographical potential included expert advice on site selection criteria that the location must meet to be suited for building utility-scale solar PV or CSP system. To analyze this, the AHP method was used to assign weights of importance to selected criteria, such as the slope of the terrain, population density of the location, proximity to power grid infrastructure, distance to road network, etc.

After the criteria selection, the next step was to identify, estimate and exclude all restricted areas from the suitable area. The Boolean binary method in the GIS environment was used to exclude all protected areas which forms the area restriction mask from the analysis. Thus, all areas that either did not meet the minimum suitable criteria or are protected were assigned the value 0 and the remaining suitable areas were assigned the value of 1. And this is the first reduction in the potential estimation of the solar PV and CSP energy output because part of the overall land area is reduced due to the exclusion mask.

Further, the remaining suitable land areas were subjected to suitability criteria. All identified suitable criteria were reclassified and assigned weights to identify areas that were either excellent or poor for the installation of each solar technology. As mentioned, the AHP was used to assign weights to all the suitable criteria of the site. The AHP is one of the most common and widely used approach and it consists of grouping criteria into a square matrix and performing a pair-wise comparison to place level and scale of importance on each suitable criterion. Based on the pair-wise matrix, a general percentage score was computed and assigned to each criterion which is also reflective of the level of importance assigned to the criteria. For this study, 5 suitability criteria were

used and then based on level of importance or prioritization, were reclassified into 5 classes with class 1 being identifies as (excellent) location, class 2 (very good), class 3 (good), class 4 (moderate), and class 5 (poor). The geographical potential maps for both technologies are shown in Figure 3.

Estimation of the technical potential

After identification of suitable locations for the solar PV and CSP systems is the estimation of the technical potential. The technical potential basically refers to the amount of electric energy that can be generated given the geographical potential by the selected solar technology over an estimated period of time. In this study, the maximum energy output per grid cell is estimated per year (GWh/year), see Figure 4. Basic methods have been proposed in many studies to estimate the technical potential or electrical potential of solar PV and CSP systems of a location [7,63,74,86,87]. For this study, the equation used by IRENA [22,63,64] and Yushchenko et al. [7] was adopted and it is given as:

$$E_i = \text{GHI}_i(\text{or } \text{DNI}_i) * \text{Efficiency} * \text{PR}_{\text{pv (for PV systems)}} * \left[\frac{\text{AvailArea}_i}{\text{SpacingFactor}} \right] \quad (4)$$

Where, E_i refers to the technical solar energy potential to be generated per year (GWh/y), i represents the grid cell value per km^2 . GHI and DNI are the solar irradiation values of the location for either PV or CSP system, respectively. $\text{PR}_{\text{pv (for PV systems)}}$ was only included in estimating the technical energy output for the solar PV technology. AvailArea_i represents the geographical potential (suitable areas) and the spacingfactor is the ratio of total land requirements to the surface of the solar panels or CSP collectors [7]. *Efficiency* as used here refers to the module efficiency for the PV or the CSP technology while PR_{pv} is the performance ratio of the PV module. The selected module specifications are based on expert advice from reviewed literature [7,63,74,86,87]. The module specification assumptions used in this study for the 2 selected solar energy technologies is shown in Table 6.

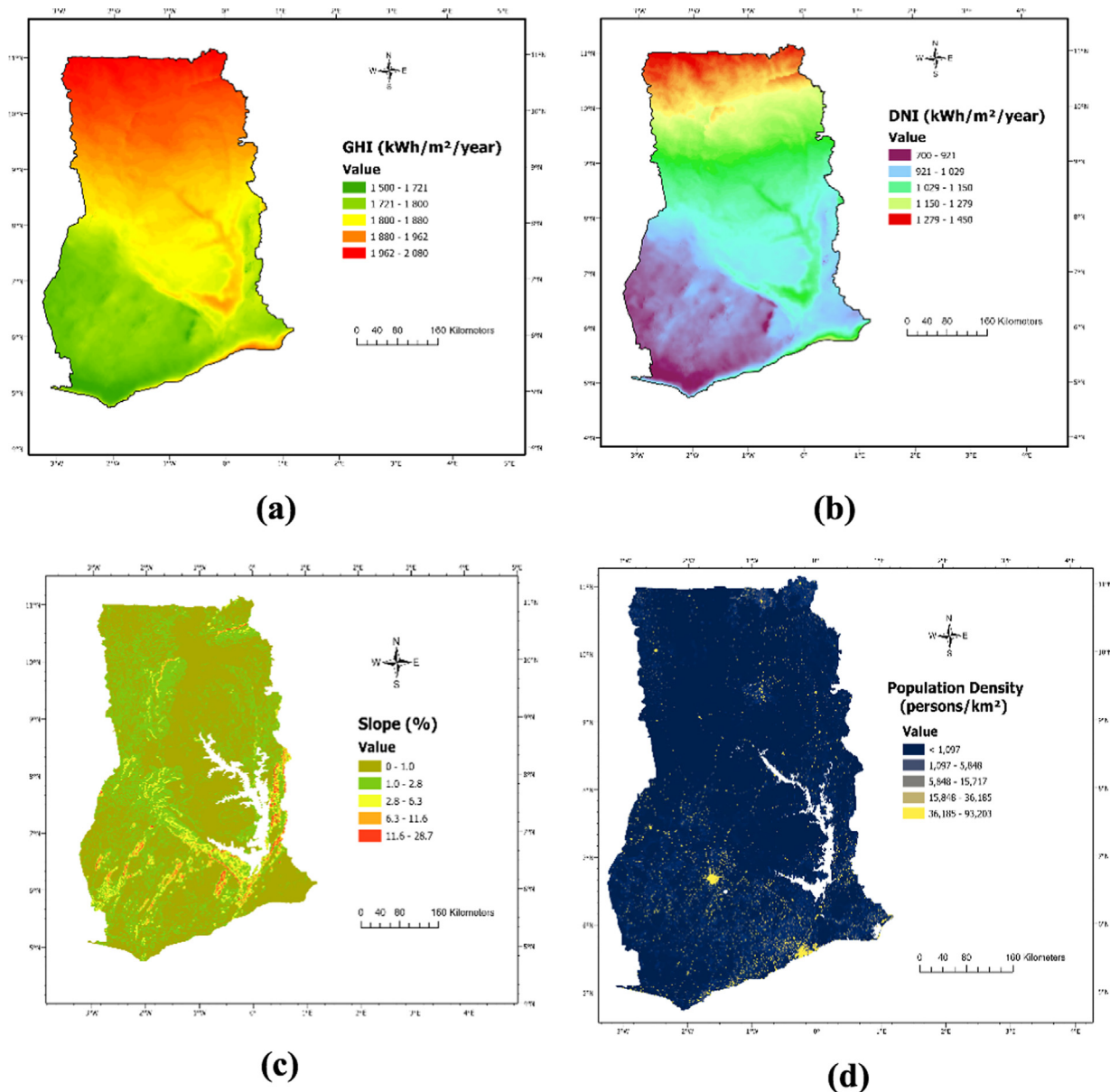


Figure 2. Input raster data for computing geographical potential. (a): Global Horizontal Irradiance, (b): Direct Normal Irradiance, (c): Slope, (d): Population Density, (e): Grid Network, (f): Roads, (g): Rivers, (h): Area restriction mask.

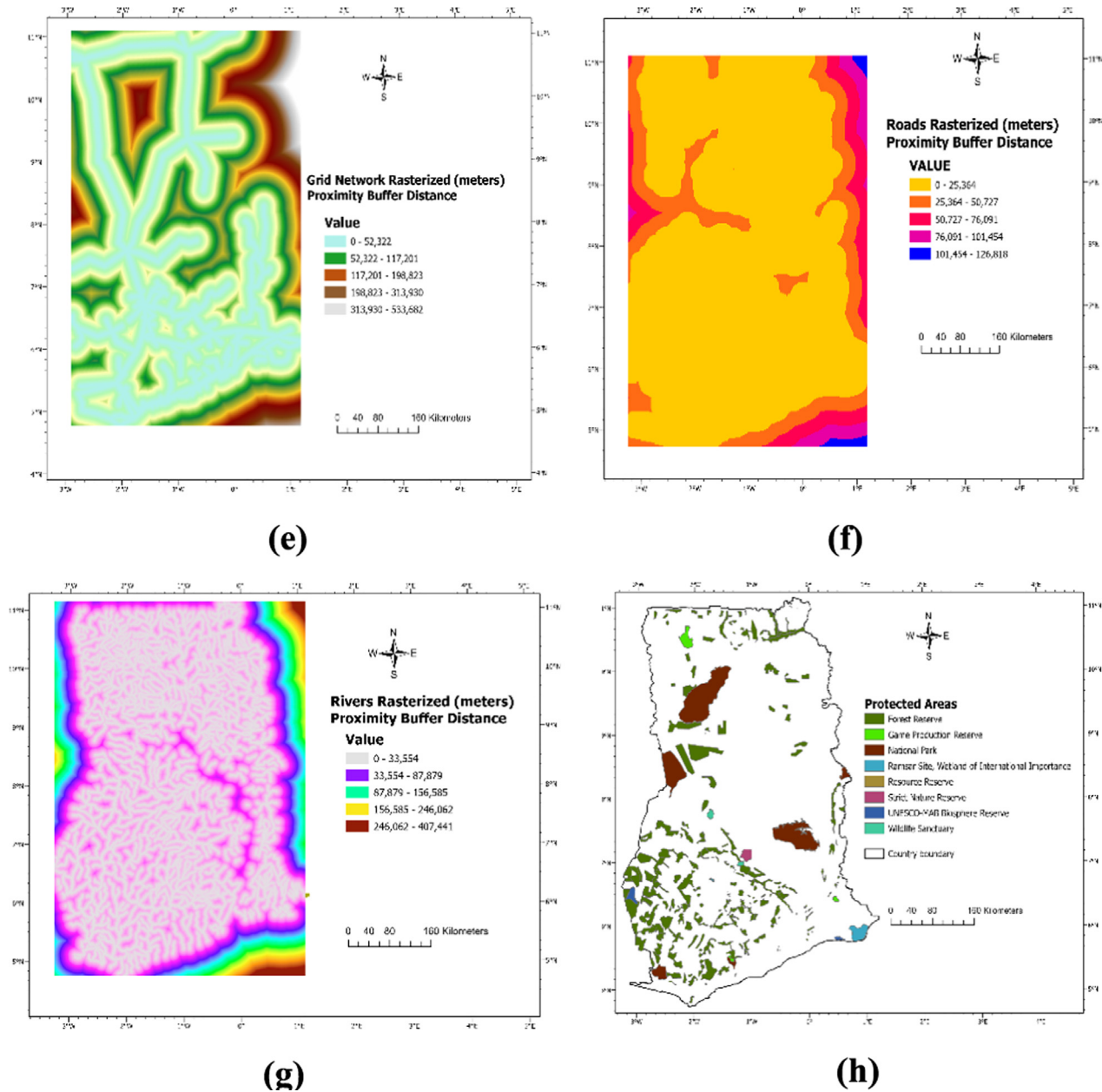


Fig. 2 (continued)

Estimation of the economic potential

The economic potential is evaluated by estimating the LCOE for utility-scale solar PV and CSP systems. As mentioned, the LCOE is used as a measure to compare the lifetime costs of generating electricity from different generation technologies [59,88,89] which includes components like the initial capital costs, operations and maintenance costs (O&M), discount rate, etc. Just like the technical potential, the LCOE is estimated based on the energy output generated per grid cell, i.e., km². The formula used for estimating the LCOE of the solar energy technologies is given as [17,61,90]:

$$LCOE_i = \left[\sum_{t=1}^n \frac{I_{ti} + M_{ti}}{(1 + r_i)^t} \right] / \left[\sum_{t=1}^n \frac{E_{ti}}{(1 + r_i)^t} \right] \quad (5)$$

Where: $LCOE_i$ refers to the average lifetime levelized cost of electricity generation per grid cell, I_{ti} is initial capital costs in the year t per grid cell, M_{ti} is the O&M costs in the year t per grid cell, E_{ti}

is the energy generated in the year t per cell, r_i is discount rate per cell and n represents the lifetime of the system.

The metrics used in this analysis are further elaborated in IRENA's renewable power generation costs in 2020 report [61]. It must however be mentioned that to estimate the LCOE per cell, all the costs component remained constant across grid cells except for the energy outputs that were the only variables. As projected in the report, initial capital costs of installed capacity per cell for utility-scale PV and CSP were \$883/kW and \$4581/kW, respectively. A 10% rate for both PV and CSP was used to estimate the O&M costs and at a discount rate of 7.5% for both technologies over a lifetime period of 25 years. The LCOE values were calculated and imported into the GIS environment to create visualisations of the economic potential maps (see Figure 7). It also important to emphasise that due to the evolving markets for the solar technologies, the LCOE results obtained in previous and subsequent years may differ due to variations in the costs components used in the estimations. For this assessment and as already stated, the cost val-

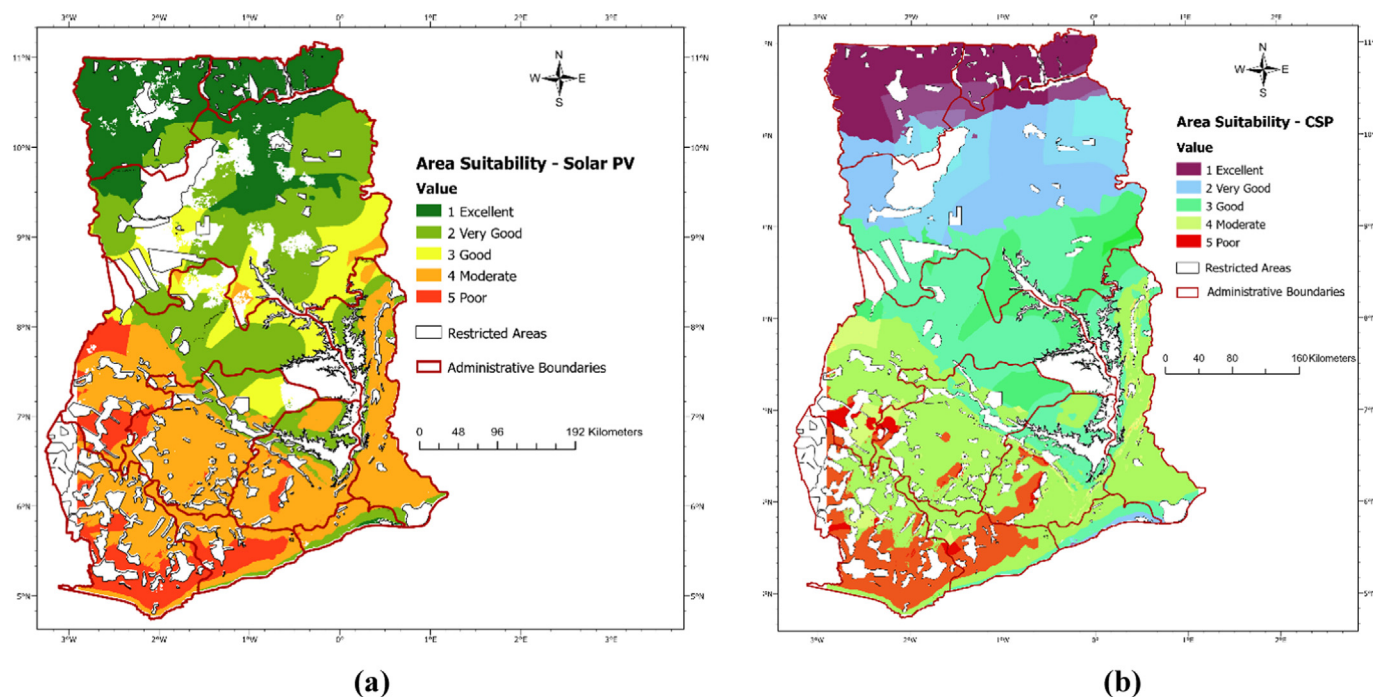


Figure 3. Area suitability Maps. (a) Area suitability for solar PV deployment, (b) Area suitability for CSP deployment.

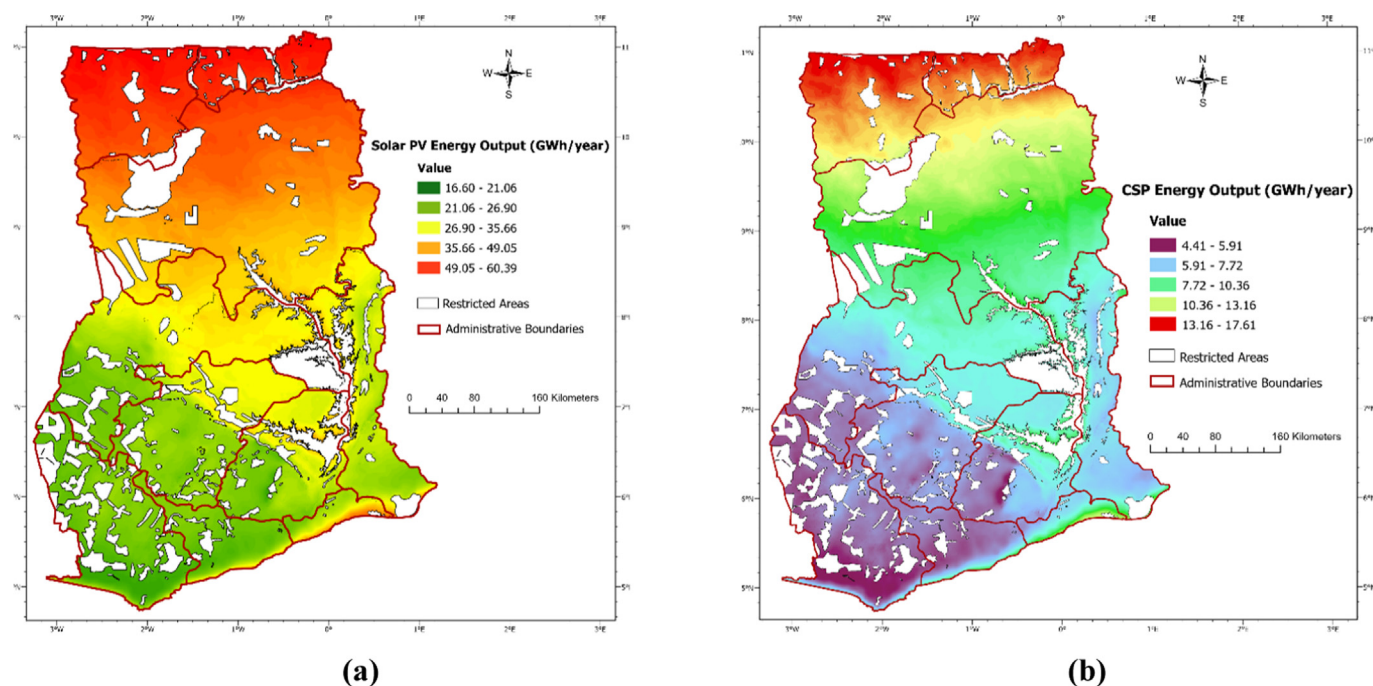


Figure 4. Maps showing maximum energy generation capacity per year per grid cell (GWh/year). (a) Utility-scale solar PV energy output, (b) CSP energy output.

Table 6

Solar PV and CSP technology specifications and assumptions used in estimating the technical solar energy potentials.

Specification	Utility-scale PV system	CSP system
Technology	Monocrystalline cells	Parabolic trough
Efficiency	Module efficiency 15–22%, PR 70–85%	System efficiency 15–21%
Spacing factor	1.4–5	3–7.5

ues used were obtained from IRENA's report due to limited data availability at the country level.

Results

In this section, results of the two energy potential categories for utility-scale solar PV and CSP technologies have been highlighted and explained. This includes results from estimation of the geographical potential, the technical potential, and the economic potential for utility-scale solar PV and CSP at the national level and within the 10 regional boundaries in Ghana. [Figure 5](#) and

Figure 6 show the geospatial models used to estimate the geographical and technical potentials, respectively. As mentioned earlier, the overarching purpose of this study is to help energy policy makers in Ghana make better decisions in their sustainable energy access planning and RE integration initiatives, and more importantly to encourage decentralization of energy access interventions at the local government level (regional level).

Geographical potential

This section provides insights into the outcome of the analysis of this study which includes site evaluation of suitable areas for development of the 2 selected solar energy technologies. From the dataset used [91], Ghana has a total land area of about 238,723 km² and about 203,795 km² (85%) area of the land is suitable for both utility-scale solar PV and CSP development. Total restricted area is about 34,928 km². Site evaluation for the 10 regions in Ghana show varying geographical potential with Northern region having the largest suitable land area (30%) for the development of both solar technologies in Ghana and with the Greater Accra region having the smallest available land (2%). Table 7 and Table 8 show the various percentage share of actual land area and suitable area both at the regional and national levels in Ghana for the two technologies

Site evaluations show varying geographical potentials across the regions in Ghana. At the regional level, the Upper West region shows the area with the highest generation capacity of about 56% (class 1), and least with potential share of 0.14% in the Greater Accra and the Volta regions for utility-scale solar PV. Nationally,

about 51% (classes 1–3) of total land area in Ghana is very ideal for development of utility-scale PV solution and only about 8% (class 5) of suitable areas show poor potential for utility-scale solar PV, see Table 7.

Also, despite the nominal DNI intensity in Ghana, suitable areas for CSP system however show great potential as shown in Table 8. For instance, about 62% (classes 1–3) of suitable areas in Ghana show promising potentials. Only about 9% (class 5) of suitable areas show poor potential, with the Volta region having the poorest technical potential per suitable area of about 0.14%. The Upper West region still has the largest CSP technical potential per suitable areas of about 56% (in class 1). See Figure 3 for geospatial suitability maps for the 2 selected technologies under research.

Technical potential

In this section, results from the technical estimation of the annual energy output generated by the 2 selected solar energy technologies have been explained. As mentioned earlier, the technical potential here refers to the amount of solar energy that the utility-scale solar PV and CSP systems can generate annually given the geographical potentials or suitable areas. It must be emphasized that, the energy output visualized in Figure 4 are based on a grid cell values where the maximum value per grid cell was used to estimate the technical potential of the location in GWh/year. But for the total capacity generated per region and nationally, sum of all grid cell values within a region was used i.e., in TWh/year.

Results of the analysis show that, a total of ~68,622 TWh/year of solar energy can be generated annually from utility-scale PV sys-

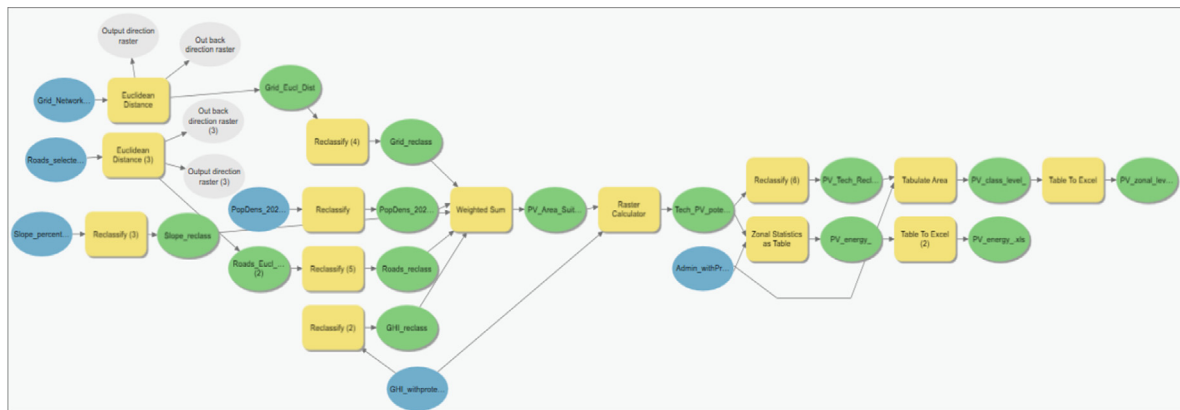


Figure 5. Geographical and technical potential estimation model for the utility-scale PV system.

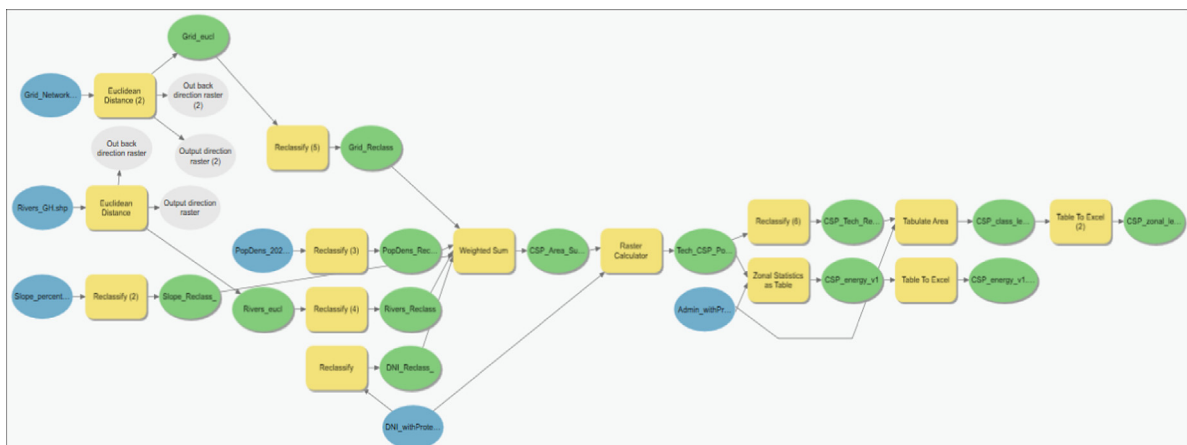


Figure 6. Geographical and technical potential estimation model for the CSP system.

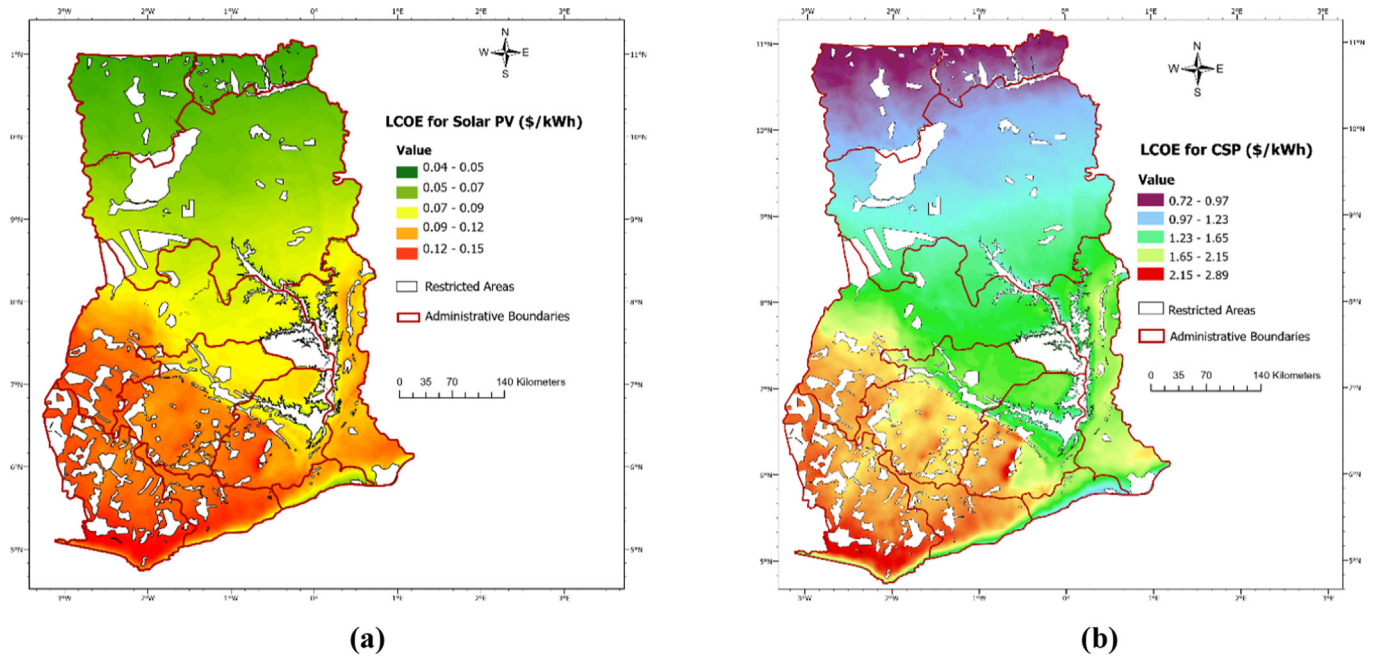


Figure 7. Maps showing the LCOE estimates per grid cell for both solar technologies. (a) LCOE estimates for Utility-scale solar PV system, (b) LCOE estimates for CSP system.

Table 7

Share of utility-scale solar PV technical potential per suitability class (%).

Region	Class	1 (Excellent)	2 (Very Good)	3 (Good)	4 (Moderate)	5 (Poor)
Ashanti		–	–	5.76%	21.07%	4.84%
Brong Ahafo		–	–	22.78%	17.30%	24.48%
Central		–	–	0.71%	6.79%	21.84%
Eastern		–	–	4.99%	14.73%	4.30%
Greater Accra		–	0.91%	2.35%	2.46%	0.14%
Northern		10.39%	68.51%	56.42%	9.74%	–
Upper East		33.27%	5.71%	–	–	–
Upper West		56.33%	24.85%	2.44%	–	–
Volta		–	0.02%	4.55%	17.36%	0.14%
Western		–	–	0.01%	10.55%	44.27%
Total		10.13%	10.11%	30.78%	41.08%	7.89%

Table 8

Share of CSP technical potential per suitability class (%).

Region	Class	1 (Excellent)	2 (Very Good)	3 (Good)	4 (Moderate)	5 (Poor)
Ashanti		–	–	10.38%	24.81%	6.07%
Brong Ahafo		–	–	31.60%	20.34%	6.87%
Central		–	–	0.90%	8.05%	21.72%
Eastern		–	–	6.94%	15.15%	11.01%
Greater Accra		0.01%	1.39%	2.44%	2.77%	–
Northern		16.43%	89.96%	38.89%	–	–
Upper East		28.06%	–	–	–	–
Upper West		55.50%	8.49%	–	–	–
Volta		–	0.16%	8.80%	18.46%	0.04%
Western		–	–	0.06%	10.41%	54.28%
Total		13.30%	19.46%	28.97%	29.53%	8.74%

tem in Ghana, see Table 9. Regionally, the Northern region generated the highest utility-scale PV energy with capacity of about 22, 203 TWh/year with a national geographical potential share of 30%, followed by the Upper West region with a technical potential of about 10,517 TWh/year with a national geographical potential share of 9%. The Greater Accra region at the national level, generated the lowest technical potential of about 1079 TWh/year given its geographical potential of 2%. At the regional level however, the Western region generated the lowest maximum capacity of ~ 28.83 GWh/year, followed by the Ashanti region with maximum capacity per grid cell of ~ 30.11 GWh/year. The CSP system generated an annual capacity of about 23,453 TWh/year in Ghana. Again,

under the criteria and assumptions used for this study, the Northern region showed the highest annual generation capacity of ~8560 TWh/year. The Greater Accra and the Central regions showed the lowest annual CSP potential of ~377 TWh/year and ~685 TWh/year, respectively, see Table 10.

Economic potential

In this study, the LCOE approach was used to estimate the economic potential for utility-scale solar PV and CSP technologies for Ghana. This approach was used to provide insights and estimation of the costs involved in the generation and integration per unit of

Table 9

Geographical, Technical and Economic potential of utility-scale solar PV system for Ghana.

Region	Actual Land Area (km ²)	Suitable Area (km ²)	National share of suitable area (%)	Share of suitable area within region (%)	PV Energy _{Min} (GWh/year)	PV Energy _{Max} (GWh/year)	PV Energy _{Mean} (GWh/year)	PV Energy _{Total} (TWh/year)	PV LCOE _{Max} (\$/kWh)	PV LCOE _{Min} (\$/kWh)	PV LCOE _{Mean} (\$/kWh)
Ashanti	24,793	20,847	10.23%	84.08%	16.90	30.11	24.12	5792.16	0.1452	0.0815	0.1018
Brong Ahafo	39,670	32,626	16.01%	82.24%	16.60	30.74	25.88	9225.66	0.1478	0.0798	0.0948
Central	9655	8777	4.31%	90.91%	17.04	29.95	21.37	2244.00	0.1440	0.0819	0.1148
Eastern	18,661	17,206	8.44%	92.21%	17.14	30.74	24.04	4232.59	0.1432	0.0798	0.1021
Greater Accra	3671	3543	1.74%	96.51%	18.66	40.73	26.45	1078.86	0.1315	0.0603	0.0928
Northern	69,037	60,238	29.56%	87.25%	23.53	58.94	34.06	22203.45	0.1043	0.0416	0.0721
Upper East	8665	7369	3.62%	85.04%	47.29	60.10	57.77	5067.24	0.0519	0.0408	0.0425
Upper West	19,466	17,370	8.52%	89.23%	30.59	60.39	52.77	10516.61	0.0802	0.0406	0.0465
Volta	20,467	18,847	9.25%	92.08%	18.11	40.11	24.58	4664.48	0.1355	0.0612	0.0998
Western	24,639	16,972	8.33%	68.88%	16.90	28.83	20.68	3597.14	0.1452	0.0851	0.1187
Total	238,723	203,795	100%	85%				68622.18			

Table 10

Geographical, Technical and Economic potential of CSP system for Ghana.

Region	Actual Land Area (km ²)	Suitable Area (km ²)	National share of suitable area (%)	Share of suitable area within region (%)	CSP Energy _{Min} (GWh/year)	CSP Energy _{Max} (GWh/year)	CSP Energy _{Mean} (GWh/year)	CSP Energy _{Total} (TWh/year)	CSP LCOE _{Max} (\$/kWh)	CSP LCOE _{Min} (\$/kWh)	CSP LCOE _{Mean} (\$/kWh)
Ashanti	24,793	20,847	10.23%	84.08%	4.92	9.08	7.00	1879.54	2.5879	1.4023	1.8189
Brong Ahafo	39,670	32,626	16.01%	82.24%	5.08	9.67	7.83	3050.29	2.5064	1.3167	1.6261
Central	9655	8777	4.31%	90.91%	5.12	11.53	6.11	685.18	2.4868	1.1043	2.0839
Eastern	18,661	17,206	8.44%	92.21%	4.41	9.67	7.07	1300.66	2.8872	1.3167	1.8009
Greater Accra	3671	3543	1.74%	96.51%	5.23	14.79	8.49	376.77	2.4345	0.8609	1.4997
Northern	69,037	60,238	29.56%	87.25%	7.36	16.40	11.19	8559.79	1.7300	0.7764	1.1379
Upper East	8665	7369	3.62%	85.04%	12.73	17.25	16.49	1520.68	1.0002	0.7381	0.7721
Upper West	19,466	17,370	8.52%	89.23%	11.47	17.61	15.63	3488.14	1.1101	0.7230	0.8146
Volta	20,467	18,847	9.25%	92.08%	4.58	12.22	7.51	1491.73	2.7800	1.0419	1.6954
Western	24,639	16,972	8.33%	68.88%	4.68	10.53	5.68	1099.85	2.7206	1.2092	2.2417
Total	238,723	203,795	100%	85.37%				23452.64			

solar energy generated by the technology. As mentioned earlier, the entire study is based on a grid cell assessment and hence the maximum values as shown on the map represents the potential LCOE estimate of the maximum energy output generated within the grid cell, i.e., per km² as visualized on the maps in Figure 7.

The LCOE for a CSP technology in Ghana is much more expensive than for the utility-scale solar PV, see Table 9 and Table 10. For example, in the Upper West; the region with the largest generation capacities for both technologies, the maximum energy generated by the PV system i.e., 60.36 GWh has LCOE of \$0.04/kWh while that of the CSP system with maximum capacity of about 17.16 GWh has unit cost or LCOE of \$0.72/kWh. From the study and as shown on the maps in Figure 7, the LCOE for the utility-scale solar PV technology ranges from about \$0.04/kWh to \$0.15/kWh, and from a minimum of \$0.73/kWh to a maximum of \$2.89/kWh for a CSP technology in Ghana.

Discussion

In this study, a combined geospatial-AHP multicriteria decision methods were applied to the case of Ghana to estimate the geographical, technical, and economic viability of utility-scale solar PV and CSP systems in the country. The GIS Boolean method was applied as the first step to delimit and delineate suitable areas from the restricted areas. Areas that were classified as restricted or delimited such as protected areas were completely excluded from the analysis. The suitability area criteria were based on extensive review of literature [7,37,43,63,64,72,74,84], etc. It must however be stated that, even though similarities exist between criteria selections, value thresholds differ slightly based on technology assumptions, geographical location, and objective of the solar project. The analysis was carried out at a grid cell resolution of 1km² per land area. The AHP [30] which is most widely used and the WSA multicriteria methods were used in the ArcGIS environment to compute the geographical and technical potentials of the solar technologies used. The values for the LCOE were however computed in excel and then imported into the GIS environment. This study is particularly relevant to Ghana's decarbonization agenda. Further, a review of the literature shows limited geospatial solar energy potential assessment in Ghana to assist policymakers make informed decisions. The only similar studies that have been carried out were done by IRENA for the Africa and the ECOWAS region [17,61,63,64] and technical potential for countries in West Africa by Yushchenko et al. [7] and not specifically on Ghana and hence the level of detail is not comparable. Results of this assessment show varying potentials for the 2 selected solar technologies, with total generation capacities of ~68,622 TWh for a utility-scale solar PV technology and ~23,453TWh for the CSP technology. The LCOE for the utility-scale solar PV technology ranges from a minimum of about \$0.04/kWh to a maximum of \$0.15/kWh, and from ~ \$0.73/kWh to ~\$2.89/kWh for the CSP technology in Ghana. The LCOE estimations are comparable as they fall within values obtained in other studies like that of [62]. Evidence from other countries have also shown that where there are right regulatory and policy frameworks in place, access to low-cost finance is available [92].

Conclusion

The techno-economic potential and site viability for utility-scale solar PV and CSP system for Ghana have been evaluated. Out of the about 238,723 km² of Ghana's land areas, about 203,795 km² (85%) available land is suitable for solar energy development. In addition, ~68,622 TWh/year of energy from utility-

scale PV can be deployed annually in Ghana, and ~23,453 TWh/year from the CSP technology. The economic potential (LCOE) for the utility-scale solar PV technology ranges from about \$0.04/kWh to \$0.15/kWh, and from about \$0.73/kWh to \$2.89/kWh for the CSP technology in Ghana. Also as already discussed, there are various factors that accounted for the unit cost of each technology. For Ghana, aside the evolving markets for the solar technologies, the overall capital costs, the O&M costs, and present discount rate, another major reason for the higher unit cost especially for the CSP technology is the low DNI intensity in the country.

Further studies are however needed to analyze the implementation potential for the selected solar technologies to understand the feasibility of implementing these projects given institutional constraints and incentives as evidence have shown that access to low-cost finance exist when the right policy framework is in place. In addition, an assessment of how to strengthen grid network connectivity to increase the uptake of solar energy would be valuable to reduce transmission and distribution losses.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Energy Commission of Ghana, Ghana – Action Plan for Sustainable Energy for All by 2030.: SE4ALL-GHANA ACTION PLAN, 2012.
- [2] World Bank, Beyond-Connections-Introducing-Multi-Tier-Framework-for-Tracking-Energy-Access-1.
- [3] World Bank, Tracking sdg7 the energy progress report full_report, 2018.
- [4] U. N. S. Division, — SDG Indicators. [Online]. Available: <https://unstats.un.org/sdgs/report/2020/goal-07/> (accessed: Jun. 14 2021).
- [5] World Bank, Sustainable Energy for All (SE4ALL), Access to electricity (% of population) - Ghana | Data. [Online]. Available: <https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=GH> (accessed: Jun. 14 2021).
- [6] N.C. Lee, V. Leal, Renew. Sustain. Energy Rev. 31 (2014) 202–220, <https://doi.org/10.1016/j.rser.2013.11.044>.
- [7] A. Yushchenko, A. de Bono, B. Chatenoux, M. Kumar Patel, N. Ray, Renew. Sustain. Energy Rev. 81 (2018) 2088–2103, <https://doi.org/10.1016/j.rser.2017.06.021>.
- [8] IEA, IEA Energy Atlas. [Online]. Available: <http://energyatlas.iea.org/#/tellmap/-1118783123/1> (accessed: Jun. 14 2021).
- [9] F.S. Javadi et al., Renew. Sustain. Energy Rev. 19 (2013) 402–416, <https://doi.org/10.1016/j.rser.2012.11.053>.
- [10] M.G. Pereira, M.A.V. Freitas, N.F. da Silva, Renew. Sustain. Energy Rev. 14 (4) (2010) 1229–1240, <https://doi.org/10.1016/j.rser.2009.12.013>.
- [11] A. Beltran, L. Laborie, P. Lanthier, S. Le Gallic (Eds.), *The Akosombo Dam and the Quest for Rural Electrification in Ghana: Creations, circulations, tensions, transitions (19th–21th C.)* / Alain Beltran, Léonard Laborie, Pierre Lanthier, Stéphanie Le Gallic (eds.). Bruxelles: P.I.E. Peter Lang, 2016. Accessed: Jun. 14 2021. [Online]. Available: <https://www.peterlang.com/view/product/62126?format=PBK>.
- [12] International Hydropower Association (IHA), Ghana - International Hydropower Association. [Online]. Available: <https://www.hydropower.org/country-profiles/ghana> (accessed: Jun. 14 2021).
- [13] Energy Commission of Ghana, 2020 ENERGY (SUPPLY AND DEMAND) OUTLOOK FOR GHANA, [Online]. Available: <http://www.energycom.gov.gh/planning/data-center/energy-outlook-for-ghana>.
- [14] Energy Commission of Ghana, 2018 Energy (Supply and Demand) Outlook for Ghana, 2018.
- [15] Energy Commission Of Ghana (EC), "National Energy Statistics 2007 - 2016," Apr. 2017.
- [16] IRENA & the German Federal Minister for Economic Cooperation and Development, *The Renewable Energy Transition in Africa: Powering Access, Resilience and Prosperity*, Mar. 2021.
- [17] IRENA, *Renewable Power Generation Costs in 2019*, 2019.
- [18] J. Hernández-Moro, J.M. Martínez-Duart, *Analytical model for solar PV and CSP electricity costs: Present LCOE values and their future evolution*, 2013. [Online]. Available: <https://reader.elsevier.com/reader/sd/pii/S136403211200696X?>

- token=19A7A04F1FBA09003DC340EFBFCADF7BB0D569B3645E2C4C164189D1B65A203FE507BB055A16155B0660456B1980C404F&originRegion=eu-west-1&originCreation=20220317131847.
- [19] Ministry of Energy, Energy Sector Recovery Program, May, 2019. [Online]. Available: http://energycom.gov.gh/files/2019%201111%20ESRP%20ESTF_Clean_v3.0redacted%20final.pdf.
 - [20] Ministry of Finance, Annual report on the management of the energy sector levies and accounts for 2019, [Online]. Available: https://mofep.gov.gh/sites/default/files/reports/economic/2019_Annual_ELSA_Report.pdf.
 - [21] A.-A. Mary, Energy 225 (2021) 120217, <https://doi.org/10.1016/j.energy.2021.120217>.
 - [22] International Renewable Energy Agency (IRENA), IRENA-IEA-ETSAP Technology Brief 1: Concentrating Solar Power (CSP), 2013.
 - [23] H.Z. Al Garmi, A. Awasthi, Appl. Energy 206 (2017) 1225–1240, <https://doi.org/10.1016/j.apenergy.2017.10.024>.
 - [24] IRENA, Ghana Renewables Readiness Assessment.
 - [25] S. Freitas, C. Catita, P. Redweik, M.C. Brito, Renew. Sustain. Energy Rev. 41 (2015) 915–931, <https://doi.org/10.1016/j.rser.2014.08.060>.
 - [26] S.D. Pohekar, M. Ramachandran, Renew. Sustain. Energy Rev. 8 (4) (2004) 365–381, <https://doi.org/10.1016/j.rser.2003.12.007>.
 - [27] J.-J. Wang, Y.-Y. Jing, C.-F. Zhang, J.-H. Zhao, Renew. Sustain. Energy Rev. 13 (9) (2009) 2263–2278, <https://doi.org/10.1016/j.rser.2009.06.021>.
 - [28] A. Kumar et al., Renew. Sustain. Energy Rev. 69 (2017) 596–609, <https://doi.org/10.1016/j.rser.2016.11.191>.
 - [29] Y. Choi, J. Suh, S.-M. Kim, Appl. Sci. 9 (9) (2019) 1960, <https://doi.org/10.3390/app9091960>.
 - [30] Thomas L. Saaty, *The analytic process: planning, priority setting, resources allocation*, 1980. [Online]. Available: <https://scholar.google.com/citations?user=mcapoueaaj&hl=en&oi=sra>.
 - [31] L.A. Zadeh, *Fuzzy sets*, 8th ed., 1965.
 - [32] P.C. Fishburn, Oper. Res. 15 (3) (1967) 537–542, <https://doi.org/10.1287/opre.15.3.537>.
 - [33] N.H. Anderson, Psychon. Sci. 8 (6) (1967) 227–228, <https://doi.org/10.3758/BF03331634>.
 - [34] J. Jensen, E. Christensen, Sci. Total Environ. 56 (1986) 265–276, [https://doi.org/10.1016/0048-9697\(86\)90331-1](https://doi.org/10.1016/0048-9697(86)90331-1).
 - [35] P.W. Bridgman, *Dimensional Analysis* (Revised edition), Yale University Press, New Haven, CT, U.S.A., 1922.
 - [36] R. Benayoun, B. Roy, B. Sussmann, *ELECTRE: Une methode pour guider le choix en presence de points de vue multiples*. Note Trav. 49, Dir. Sci., Soc. Econ. Math. Appl., Paris, 1966.
 - [37] C.-L. Hwang, K. Yoon, *Multiple attribute decision making: Methods and applications: a state the Art Survey*. Berlin [etc.], Springer, 1981.
 - [38] W. Edwards, J.R. Newman, *Multiattribute Evaluation*, Beverly Hills, Calif., London, Sage, 1982.
 - [39] J.P. Brans, P. Vincke, Manage. Sci. 31 (6) (1985) 647–656, <https://doi.org/10.1287/mnsc.31.6.647>.
 - [40] N. Balakrishnan, T. Colton, B. Everitt, W.W. Piegorsch, F. Ruggeri, J.L. Teugels (Eds.), *Wiley StatsRef: Statistics reference online*. John Wiley & Sons, Inc, Hoboken, NJ, 2014.
 - [41] T.P. Fluri, Energy Policy 37 (12) (2009) 5075–5080, <https://doi.org/10.1016/j.enpol.2009.07.017>.
 - [42] E. Zawilska, M.J. Brooks, Renewable Energy 36 (12) (2011) 3433–3438, <https://doi.org/10.1016/j.renene.2011.05.023>.
 - [43] H.E. Colak, T. Memisoglu, Y. Gercek, Renewable Energy 149 (2020) 565–576, <https://doi.org/10.1016/j.renene.2019.12.078>.
 - [44] I. Gašparović, M. Gašparović, Remote Sensing 11 (12) (2019) 1481, <https://doi.org/10.3390/rs11121481>.
 - [45] R. Djebbar, D. Belanger, D. Boutin, E. Weterings, M. Poirier, Energy Proc. 49 (2013) 2303–2312, <https://doi.org/10.1016/j.egypro.2014.03.244>.
 - [46] J.R. Janke, Renewable Energy 35 (10) (2010) 2228–2234, <https://doi.org/10.1016/j.renene.2010.03.014>.
 - [47] R. Ramanathan, L.S. Ganesh, Socio-Econ. Plann. Sci. 29 (3) (1995) 197–218, [https://doi.org/10.1016/0038-0121\(95\)00013-C](https://doi.org/10.1016/0038-0121(95)00013-C).
 - [48] M. Ibarra, I. Gherboudj, A. Al Rished, H. Ghedira, in: Abu Dhabi, United Arab Emirates, 2017, p. 160015.
 - [49] A. Sunarso, K. Ibrahim-Bathis, S.A. Murti, I. Budiarto, H.S. Ruiz, Sustainability 12 (15) (2020) 6283, <https://doi.org/10.3390/su12156283>.
 - [50] H.S. Ruiz, A. Sunarso, K. Ibrahim-Bathis, S.A. Murti, I. Budiarto, Energy Rep. 6 (2020) 3249–3263, <https://doi.org/10.1016/j.egy.2020.11.198>.
 - [51] L. Sun et al., Resour. Conserv. Recycl. 168 (2021) 105306, <https://doi.org/10.1016/j.resconrec.2020.105306>.
 - [52] A. Alami Merrouni, F. Elwali Elalaoui, A. Ghennoui, A. Mezrhah, A. Mezrhah, Sol. Energy 166 (2018) 2–12, <https://doi.org/10.1016/j.solener.2018.03.038>.
 - [53] G. Rediske, J.C.M. Siluk, L. Michels, P.D. Rigo, C.B. Rosa, G. Cugler, Energy 197 (2020) 117167, <https://doi.org/10.1016/j.energy.2020.117167>.
 - [54] M. Huda, K. Okajima, K. Suzuki, Energy Proc. 142 (2017) 2554–2560, <https://doi.org/10.1016/j.egypro.2017.12.091>.
 - [55] S. Ziuku, L. Seyitini, B. Mapurisa, D. Chikodzi, K. van Kuijk, Energy Sustain. Dev. 23 (2014) 220–227, <https://doi.org/10.1016/j.esd.2014.07.006>.
 - [56] U. Tempa, J.G. Singh, in: International Conference on Electrical and Electronics Engineering - ICEE-2020: February 14–15, 2020, Gorakhpur, India, 2020, pp. 705–710.
 - [57] F. Kemausuor, E. Adkins, I. Adu-Poku, A. Brew-Hammond, V. Modi, Energy Sustain. Dev. 19 (2014) 92–101, <https://doi.org/10.1016/j.esd.2013.12.009>.
 - [58] F. Kemausuor, G.Y. Obeng, A. Brew-Hammond, A. Duker, Renew. Sustain. Energy Rev. 15 (9) (2011) 5143–5154, <https://doi.org/10.1016/j.rser.2011.07.041>.
 - [59] S. Raikar, S. Adamson, in: *Renewable Energy Finance*, Elsevier, 2020, pp. 185–220.
 - [60] C. Tjengdrawira, Richter, Mauricio, et al., Review and Gap Analyses of Technical Assumptions in PV Electricity Cost, Report on Current Practices in How Technical Assumptions Are Accounted in PV Investment Cost Calculation, 2016.
 - [61] IRENA, Renewable Power Generation Costs 2020, [Online]. Available: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2021/Jun/IRENA_Power_Generation_Costs_2020.pdf.
 - [62] E.B. Agyekum, V.I. Velkin, Sustainable Energy Technol. Assess. 40 (2020) 100763, <https://doi.org/10.1016/j.seta.2020.100763>.
 - [63] International Renewable Energy Agency (IRENA), Estimating the Renewable Energy Potential in Africa: A GIS-based approach.
 - [64] International Renewable Energy Agency (IRENA), Unleashing the solar potential in ECOWAS: Seeking areas of opportunity.
 - [65] The World Factbook, *Ghana - The World Factbook*. [Online]. Available: <https://www.cia.gov/the-world-factbook/countries/ghana/#geography> (accessed: Dec. 20 2021).
 - [66] Worldfacts, *Geography of Ghana - Geographical Regions, Rivers and Lakes, Climate*. [Online]. Available: <http://worldfacts.us/Ghana-geography.htm> (accessed: Dec. 20 2021).
 - [67] Ghana Statistical Service, 2021 Population and Housing Census. Provisional Results Press Release, [Online]. Available: <https://statsghana.gov.gh/gssmain/storage/img/infobank/2021%20PHC%20Provisional%20Results%20Press%20Release.pdf>.
 - [68] Ghana Permanent Mission to the United Nations, *Map & Regions in Ghana - Ghana Permanent Mission to the United Nations*. [Online]. Available: <https://www.ghanamissionun.org/map-regions-in-ghana/> (accessed: Dec. 20 2021).
 - [69] O. Marinoni, Comput. Geosci. 30 (6) (2004) 637–646, <https://doi.org/10.1016/j.cageo.2004.03.010>.
 - [70] Edward D. Wagner, Public key infrastructure (PKI) and virtual private network (VPN) compared using an utility function and the analytic hierarchy process (AHP), 2002. [Online]. Available: <https://vtechworks.lib.vt.edu/bitstream/handle/10919/32685/thesiswagner.pdf?sequence=1&isAllowed=y>.
 - [71] T. Huang, S. Wang, Q. Yang, J. Li, *A GIS-based assessment of large-scale PV potential in China*, 2018. Accessed: Jun. 14 2021. [Online]. Available: <https://reader.elsevier.com/reader/sd/pii/S1876610218306702?token=39DFDB0904A4FDF2A86FDE5B0F3F98B28FEC330B4B1A221374A739BD4ECAB5469B254C5077B22BC177913B7E8FD5166A>.
 - [72] E. Noorollahi, D. Fadaei, M. Akbarpour Shirazi, S. Ghodsipour, Energies 9 (8) (2016) 643, <https://doi.org/10.3390/en9080643>.
 - [73] T. Ramachandra, Solar energy potential assessment using GIS, *undefined*, 2007.
 - [74] G. Ghasemi, Y. Noorollahi, H. Alavi, M. Marzbani, M. Shahbazi, Renewable Energy 138 (2019) 1250–1261, <https://doi.org/10.1016/j.renene.2019.02.068>.
 - [75] World Bank Group, *Global Solar Atlas*. [Online]. Available: <https://globalsolaratlas.info/map?c=-7.966758,-1.0327157&r=GHA> (accessed: Jun. 14 2021).
 - [76] Global Terrain Slope and Aspect Data, *IIASA - Land Use Change and Agriculture Program*. [Online]. Available: <http://webarchive.iiasa.ac.at/Research/LUC/External-World-soil-database/HTML/global-terrain-slope-download.html?sb=7> (accessed: Jun. 14 2021).
 - [77] Worldpop, *WorldPop:: Population Counts*. [Online]. Available: <https://www.worldpop.org/geodata/summary?id=139> (accessed: Dec. 9 2021).
 - [78] World Bank, *Ghana Electricity Transmission Network: Africa Infrastructure Country Diagnostics (AICD)*. [Online]. Available: <https://datacatalog.worldbank.org/dataset/ghana-electricity-transmission-network> (accessed: Jun. 14 2021).
 - [79] C. Hernández, R. Barraza, A. Saez, M. Ibarra, D. Estay, Energies 13 (9) (2020) 2131, <https://doi.org/10.3390/en13092131>.
 - [80] M. Balghouthi, S.E. Trabelsi, M.B. Amara, A.B.H. Ali, A. Guizani, Renew. Sustain. Energy Rev. 56 (2016) 1227–1248, <https://doi.org/10.1016/j.rser.2015.12.052>.
 - [81] L. Dawson, P. Schlyter, Energy Policy 47 (2012) 91–101, <https://doi.org/10.1016/j.enpol.2012.04.025>.
 - [82] C. Sharma, A.K. Sharma, S.C. Mullick, T.C. Kandpal, Renew. Sustain. Energy Rev. 42 (2015) 902–912, <https://doi.org/10.1016/j.rser.2014.10.059>.
 - [83] World Bank, *Ghana Rivers*. [Online]. Available: <https://datacatalog.worldbank.org/dataset/ghana-rivers> (accessed: Jun. 14 2021).
 - [84] M. Uyan, Renew. Sustain. Energy Rev. 28 (2013) 11–17, <https://doi.org/10.1016/j.rser.2013.07.042>.
 - [85] UNEP-WCMC and IUCN, *Protected Planet: The World Database: Protected Areas (WDA)*. Cambridge, UK, 2021. [Online]. Available: www.protectedplanet.net.
 - [86] M. Šúri, T.A. Huld, E.D. Dunlop, H.A. Ossensbrink, Sol. Energy 81 (10) (2007) 1295–1305, <https://doi.org/10.1016/j.solener.2006.12.007>.
 - [87] Y. Sun, A. Hof, R. Wang, J. Liu, Y. Lin, D. Yang, Energy Policy 58 (2013) 248–259, <https://doi.org/10.1016/j.enpol.2013.03.002>.
 - [88] X. Ouyang, B. Lin, Energy Policy 70 (2014) 64–73, <https://doi.org/10.1016/j.enpol.2014.03.030>.
 - [89] W. Short, D.J. Packey, T. Holt, *A manual for the economic evaluation of energy efficiency and renewable energy technologies*, 1995.
 - [90] International Renewable Energy Agency (IRENA), IRENA Cost and Competitive Indicators: Rooftop solar PV.
 - [91] Ghana Open Data Initiative, *Shapefiles of all Regions in Ghana 2010 (10 regions) | Ghana Open Data Initiative*. [Online]. Available: <https://data.gov.gh/dataset/shapefiles-all-regions-ghana-2010-10-regions> (accessed: Jun. 14 2021).
 - [92] International Renewable Energy Agency (IRENA), Solar PV in Africa: Costs and Markets.